

PART 1: Conditional probability graphs

We can use the full joint probability tables to answer any question we may have about a set of random variables; however, as the number of variables increases, the size of the joint probability table increases exponentially. Thankfully, we can use the independence and conditional independence relationships between variables to design a compact representation that greatly reduces the number of probabilities needed to define the full joint probability distribution. These structures are called *Bayesian networks* and are typically represented as directed, acyclic graphs $G = (V, E)$ where V is the set of random variables and $E \subseteq V \times V$ is the set of directed edges. Random variables are denoted by capital letters (e.g., X) and lower-case denotes the truth assignment of a variable (e.g., x or $\neg x$). Edge $(Y, X) \in E$ indicates that Y is the *parent* of X and we can denote the set of parents of X as $parents(X)$. The parents of X are the only variables that *directly* affect the probability of X . This means that given $parents(X)$, there is no other information that is useful to estimate $P(X)$, formally: $P(X | parents(X)) = P(X | V_1, V_2, \dots, V_n)$ for all random variables $V_i \in V$.

In order to compute the full joint distribution using a Bayesian network, we need to find the product of all the conditional probabilities. We can do this using the following equation:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | parents(X_i)).$$

It is clear that if X_i has no parents, its “conditional” probability is simply $P(X_i)$.

Where we are making a query about a variable but we don’t have full knowledge about its parents, we must sum over the hidden variables. This can be done in the following manner:

$$P(X) = \sum_{\mathbf{y}} P(X, \mathbf{y}),$$

This technique is known as *marginalisation* and here, $\mathbf{Y}$ is the vector of all hidden variables we wish to sum over. For example, in the case where there is one hidden variable Y , we have,

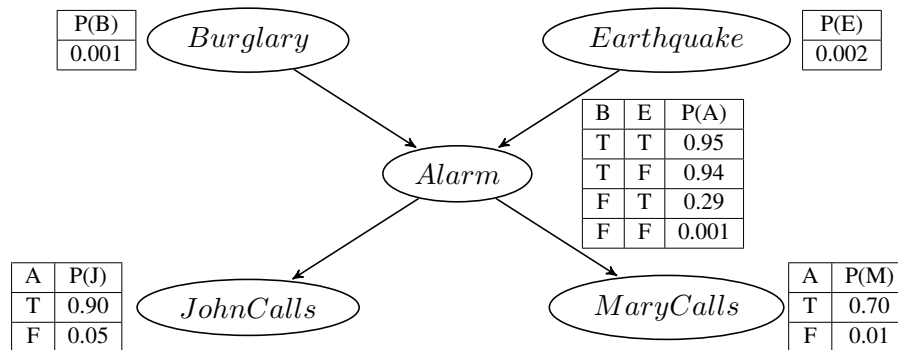
$$P(X) = P(X, y) + P(X, \neg y);$$

in the case of two hidden variables Y_1 and Y_2 we have,

$$P(X) = P(X, y_1, y_2) + P(X, y_1, \neg y_2) + P(X, \neg y_1, y_2) + P(X, \neg y_1, \neg y_2);$$

and so on. In general, we will have the sum of 2^n terms, where n is the number of hidden variables being summed over. Each of these terms will represent one of the possible truth assignments for all of the hidden variables.

Example: Consider the following Bayesian network:



- (a) Give an expression for the probability $P(j, m, a, \neg e, \neg b)$.

Solution: Here, we need to find conditional probabilities for each truth assignment of the variables, based on the structure of the graphical model given. So, j depends on a ; m depends on a ; a depends on both e and b ; and, b and e don't depend on any other variables. The total probability for the entry in the full joint table represented by $P(j, m, a, \neg e, \neg b)$ is the product of all of these conditional probabilities.

Therefore an expression for the above probability would be:

$$P(j | a) \times P(m | a) \times P(a | \neg e, \neg b) \times P(\neg e) \times P(\neg b)$$

- (b) Compute the probability $P(j, m, a, \neg e, \neg b)$.

Solution: Having worked out an expression for the probability in the previous question, computing the value is a simple matter of plugging in the correct values - remembering that $P(\neg a) = 1 - P(a)$.

Therefore, the value for the above probability is:

$$\begin{aligned}
 P(j, m, a, \neg e, \neg b) &= P(j | a) \times P(m | a) \times P(a | \neg e, \neg b) \times P(\neg e) \times P(\neg b) \\
 &= P(j | a) \times P(m | a) \times P(a | \neg e, \neg b) \times (1 - P(e)) \times (1 - P(b)) \\
 &= 0.90 \times 0.70 \times 0.001 \times (1 - 0.002) \times (1 - 0.001) \\
 &= 0.90 \times 0.70 \times 0.001 \times 0.998 \times 0.999 \\
 &= 0.0006281113
 \end{aligned}$$

- (c) Give an expression for the probability $P(j, \neg m, \neg a, b)$.

Solution: Here, we are missing one of the truth assignment to the variable e , so we must marginalise over that variable by summing up all entries in the full joint table corresponding to all of the other truth assignments and when e is both true *and* when it is false.

Therefore, an expression for the above probability is:

$$\begin{aligned}
 P(j, \neg m, \neg a, b) &= \sum_e P(j, \neg m, \neg a, b, e) \\
 &= \sum_e P(j | \neg a) \times P(\neg m | \neg a) \times P(\neg a | b, e) \times P(b) \times P(e) \\
 &= P(j | \neg a) \times P(\neg m | \neg a) \times P(\neg a | b, e) \times P(b) \times P(e) \\
 &\quad + P(j | \neg a) \times P(\neg m | \neg a) \times P(\neg a | b, \neg e) \times P(b) \times P(\neg e)
 \end{aligned}$$

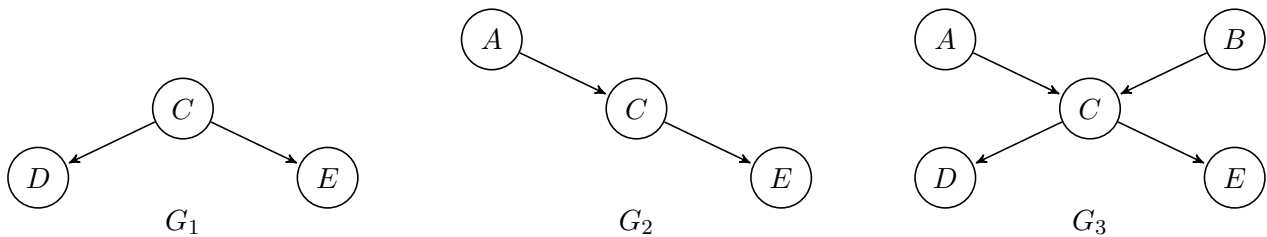
- (d) Compute the probability $P(j, \neg m, \neg a, b)$.

Solution: Just plug the values in as you did before:

$$\begin{aligned}
 P(j, \neg m, \neg a, b) &= P(j \mid \neg a) \times (1 - P(m \mid \neg a)) \times (1 - P(a \mid b, e)) \times P(b) \times P(e) \\
 &\quad + P(j \mid \neg a) \times (1 - P(m \mid \neg a)) \times (1 - P(a \mid b, \neg e)) \times P(b) \times (1 - P(e)) \\
 &= 0.05 \times 0.99 \times 0.05 \times 0.001 \times 0.002 \\
 &\quad + 0.05 \times 0.99 \times 0.06 \times 0.001 \times 0.998 \\
 &= 0.00000296901
 \end{aligned}$$

Questions:

Consider the following directed graphical models:



Give expressions for the following probabilities using only terms of the form $P(X \mid \text{parents}(X))$:

(a) with respect to graph G_1 ;

i. $P(c, d, e)$

Solution:

$$P(d \mid c) \times P(e \mid c) \times P(c)$$

ii. $P(d, e \mid c)$

Solution: Your turn!

(b) with respect to graph G_2 ;

i. $P(a, c, e)$

Solution: Your turn!

ii. $P(a, e \mid c)$

Solution:

$$\begin{aligned}
 P(a, e \mid c) &= \frac{P(a, e, c)}{P(c)} \\
 &= \frac{P(a) \times P(c \mid a) \times P(e \mid c)}{P(c)}
 \end{aligned}$$

Here, we have only the conditional probability table for C , which depends on A , so we must marginalise over A to obtain $P(c)$.

$$\frac{P(a) \times P(c \mid a) \times P(e \mid c)}{P(c)} = \frac{P(a) \times P(c \mid a) \times P(e \mid c)}{\sum_a P(c \mid a) \times P(a)} = \frac{P(a) \times P(c \mid a) \times P(e \mid c)}{P(c \mid a) \times P(a) + \dots}$$

(c) with respect to graph G_3 .

i. $P(a, \neg b, c, d, \neg e)$

Solution: Your turn!

ii. $P(a \mid \neg d, e)$

Solution:

Using conditional probability rule:

$$P(a \mid \neg d, e) = \frac{P(a, \neg d, e)}{P(\neg d, e)} = \alpha P(a, \neg d, e),$$

where α is the normalisation constant $\frac{1}{P(\neg d, e)}$.

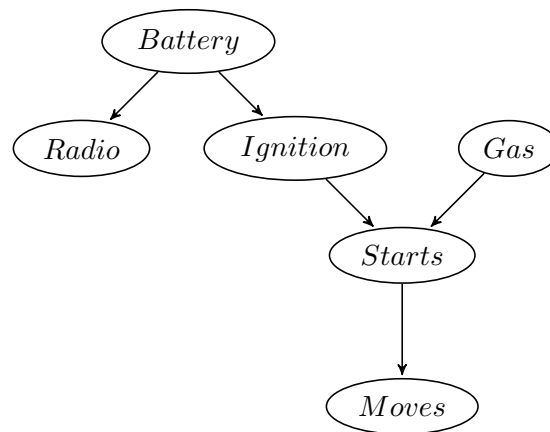
Now, because d and e are conditional on c and c is conditional on b , we have to sum up over the hidden variables b and c .

So we have,

$$\begin{aligned} P(a \mid \neg d, e) &= \alpha P(a, \neg d, e) \\ &= \alpha \sum_{b,c} P(a, \neg d, e, b, c) \\ &= \alpha \sum_{b,c} P(a) \times P(b) \times P(c \mid a, b) \times P(\neg d \mid c) \times \dots \\ &= \alpha [P(a) \times P(b) \times P(c \mid a, b) \times P(\neg d \mid c) \times P(e \mid c) \\ &\quad + P(a) \times P(b) \times P(\neg c \mid a, b) \times P(\neg d \mid \neg c) \times P(e \mid \neg c) \\ &\quad + \dots \\ &\quad + P(a) \times P(\neg b) \times P(\neg c \mid a, \neg b) \times P(\neg d \mid \neg c) \times P(e \mid \neg c)] \end{aligned}$$

PART 2: Car diagnosis

Consider the network for car diagnosis shown in the figure below:



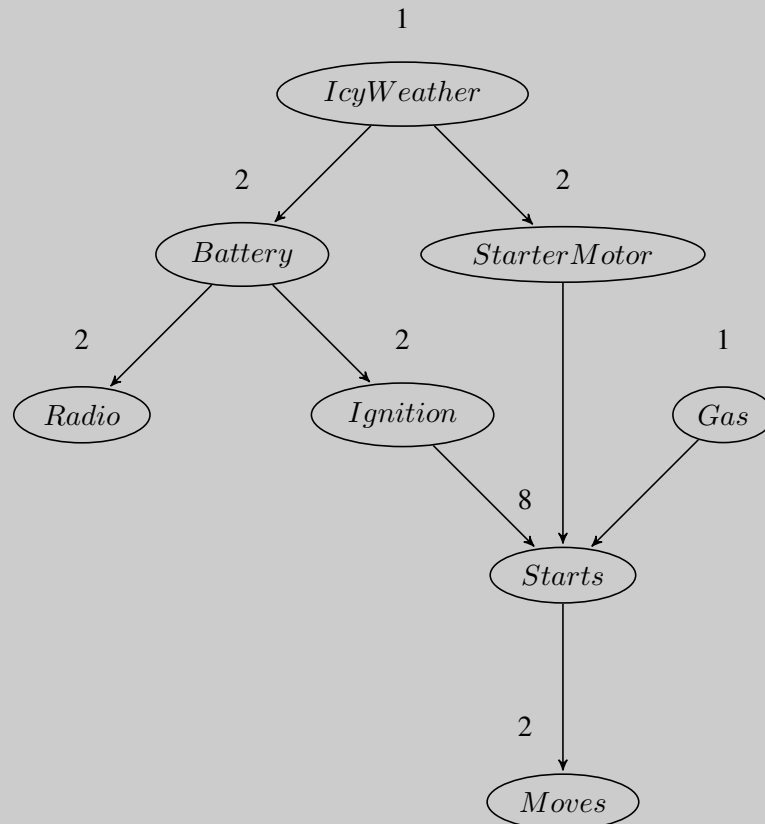
- (a) Extend the network with the Boolean variables *IcyWeather* and *StarterMotor*, by reasoning how components might affect each other. Being a design/modelling process, there may be many correct answers, just state your assumptions.

Solution:

Adding variables to an existing net can be done in two ways. Formally speaking, one should insert the variable ordering and rerun the network construction process from the point where the first new variable appears. Informally speaking, one never really builds a network by a strict ordering. Instead, one asks what variables are direct causes or influences on what other ones, and builds local parent/child graphs

that way. It is usually very easy to identify where in such a structure the new variable goes, but one must be very careful to check for possible induced dependencies downstream.

IcyWeather is not caused by any of the car-related variables, so needs no parents. It directly affects the battery and starter motor. *StarterMotor* is an additional precondition for *Starts*. The new network is shown below:



(b) Give reasonable conditional probability tables for all the nodes.

Solution: (Note: this is a partial solution)

1. A reasonable prior for *IcyWeather* might be 0.05 (perhaps, depending on the location and season).
2. $P(\text{Battery} \mid \text{IcyWeather}) = 0.95$,
 $P(\text{Battery} \mid \neg \text{IcyWeather}) = 0.997$
3. $P(\text{StarterMotor} \mid \text{IcyWeather}) = \dots$,
 $P(\text{StarterMotor} \mid \neg \text{IcyWeather}) = \dots$
4. $P(\text{Radio} \mid \text{Battery}) = \dots$,
 $P(\text{Radio} \mid \neg \text{Battery}) = \dots$
5. $P(\text{Ignition} \mid \text{Battery}) = 0.998$,
 $P(\text{Ignition} \mid \neg \text{Battery}) = 0.01$
6. $P(\text{Gas}) = 0.995$
7. $P(\text{Starts} \mid \text{Ignition}, \text{StarterMotor}, \text{Gas}) = \dots$
8. $P(\text{Moves} \mid \text{Starts}) = \dots$

There are more entries in the full CPT, however they have not been listed here for space reasons.

(c) How many independent values are contained in the joint probability distribution for eight Boolean nodes, assuming that no conditional independence relations are known to hold among them?

Solution:

With 8 Boolean variables, the joint has $2^8 - 1 = 255$ independent entries. In general there are 2^n entries in a joint probability distribution; however, only $2^n - 1$ *independent* entries. If the probabilities of all of the independent entries are summed up, the probability of the final entry must be $1 - \sum P(\text{independent})$, making its value *dependent* on the rest.

- (d) How many independent probability values do your network tables contain?

Solution: Your turn!

- (e) **(optional)** The conditional distribution for *Starts* could be described as a **noisy-AND** distribution. Define this family in general and relate it to the **noisy-OR** distribution.

Solution: (Note: this is a partial solution)

The CPT for *Starts* describes a set of nearly necessary conditions that are together almost sufficient. That is, all the entries are nearly zero except for the entry where all conditions are true. That entry will be not quite 1 (because there is always some other possible fault that we didn't think of), but as we add more conditions it gets closer to 1. If we add a *Leak* node as an extra parent, then the probability is exactly 1 when all parents are true. We can relate noisy-AND to noisy-OR using de Morgan's rule: $A \wedge B \equiv \neg(\neg A \vee \neg B)$. That is, noisy-AND is the same as noisy-OR except that the polarities of the parent and child variables are reversed. In the noisy-OR case, we have

$$P(Y = \text{true} \mid x_1, \dots, x_k) = 1 - \prod_{\{i: x_i = \text{true}\}} q_i \quad (1)$$

where q_i is the probability that the *presence* of the i th parent *fails* to cause the child to be *true*. In the noisy-AND case, we can write

$$P(Y = \text{true} \mid x_1, \dots, x_k) = 1 - \dots \quad (2)$$

where r_i is the probability that the *absence* of the i th parent *fails* to cause the child to be *false* (e.g., it is magically bypassed by some other mechanism).

- (f) Reconstruct, showing your workings, five entries of the full joint probability distribution (e.g., reconstruct $P(\neg iW, b, r, i, g, s, sM, \neg m)$ or $P(iW, \neg b, r, i, \neg g, \neg s, sM, m)$, or any other variable assignment).

Solution: (Note: this is a partial solution)

Here is one example. The probabilities are taken from question 1(b).

$$\begin{aligned} P(\neg iW, b, r, i, g, s, sM, \neg m) &= \\ P(\neg iW) \cdot P(b \mid \neg iW) \cdot P(r \mid b) \cdot P(i \mid b) \cdot P(g) \cdot P(s \mid i, sM, g) \cdot \dots &= \\ (1 - P(iW)) \cdot P(b \mid \neg iW) \cdot P(r \mid b) \cdot P(i \mid b) \cdot \dots &= \\ 0.95 \cdot 0.997 \cdot 0.9999 \cdot 0.998 \cdot 0.995 \cdot 0.9999 \cdot 0.999 \cdot 0.002 &= 0.00188 \end{aligned}$$

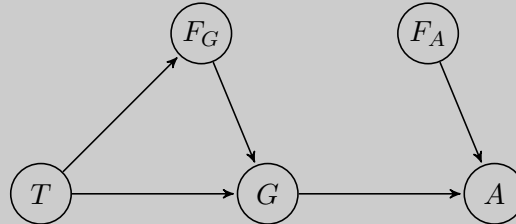
PART 3: Power station diagnosis

In your local nuclear power station, there is an alarm that senses when a temperature gauge exceeds a given threshold. The gauge measures the temperature of the core. Consider the Boolean variable A (alarm sounds), F_A (alarm is faulty), and F_G (gauge is faulty) and the multivalued nodes G (gauge reading) and T (actual core temperature).

- (a) Draw a Bayesian network for this domain, given that the gauge is more likely to fail when the core temperature gets too high.

Solution:

A suitable network is shown in the figure below. The key aspects are: the failure nodes are parents of the sensor nodes, and the temperature node is a parent of both the gauge and the gauge failure node. It is exactly this kind of correlation that makes it difficult for humans to understand what is happening in complex systems with unreliable sensors.



- (b) Is your network a polytree?

Solution: Your turn!

- (c) Suppose there are just two possible actual and measured temperatures, normal and high; the probability that the gauge gives the correct temperature is x when it is working, but y when it is faulty. Give the conditional probability table associated with G .

Solution: (Note: this is a partial solution)

The CPT for G is shown below. The wording of the question is a little tricky because x and y are defined in terms of “incorrect” rather than “correct”.

	$T = Normal$		$T = High$	
	F_G	$\neg F_G$	F_G	$\neg F_G$
$G = High$	$1 - y$	$1 - x$	...	...
$G = Normal$	y	x	...	...

- (d) Suppose the alarm works correctly unless it is faulty, in which case it never sounds. Give the conditional probability table associated with A .

Solution: (Note: this is a partial solution)

The alarm works correctly unless it is faulty. So, if the alarm is not faulty, then the alarm works correctly, meaning that it sounds only if gauge reading is high.

The CPT for A is as follows:

	$G = Normal$		$G = High$	
	F_A	$\neg F_A$	F_A	$\neg F_A$
A	0	0	0	1
$\neg A$	...	...	...	...

- (e) (optional) Suppose the alarm and gauge are working and the alarm sounds. Using the information from the previous questions, calculate an expression for the probability that the temperature of the core is too high, in terms of the various conditional probabilities in the network.

Solution: (Note: this is a partial solution)

The assumption says that the alarm and gauge are working, which means that $F_A = \neg f_a$ and $F_G = \neg f_g$.

Also, we have that the alarm sounds ($A = a$). As we are asked the probability of the temperature being too high ($T = t$), we can formulate the query as:

$$P(t \mid a, \neg f_a, \neg f_g)$$

Which we would have to evaluate for the values of G (the remaining variable in our network). However, since we are using the assumptions above (also from parts (c) and (d)) we can deduce the following: Since the alarm sounds (a) and the alarm is not faulty ($\neg f_a$), we can infer by the table above that for this to be possible, the gauge reading must be high ($G = g$). Hence, we can rewrite the above expression as:

$$P(t \mid a, \neg f_a, g, \neg f_g) = \dots$$

Your turn!

PART 4: Bayesian Networks via pgmpy

Finally, let's get our hands dirty with Bayesian Networks in Python!

One can build and do inference with Bayesian Networks in Python using the **pgmpy** package!

So, head to **pgmpy** home page and:

- (a) Install and set-up the package in your machine.
- (b) Understand and run the **Earthquake example** in the documentation.
- (c) Use the system to perform and validate some of the inferences you did in Part 1 for this domain.
- (d) **Enjoy & learn Bayesian Networks with Python!** Make sure you understand what the code is doing and that it does not appear as “magic” to you!

End of tutorial worksheet